Mothur v.1.48.3

Tegu Project(93 samples & 2 extraction blanks & 1 pcr blank/192 files)

File Download and Set-up

On My Passport in "Tegu_16S2":

- tegu_fastq_GZ2
  - *Downloaded sequences from oit.utk.edu/hpsc (Files are in .gz format.)*
- tegu_fastq_extracted2
  - *Extracted files from .gz to .fastq format using 7-zip*
  - *Files are still in folders and will need to be taken out for the working file!!*
- tegu_wd
  - *This is the working file.*
  - *Files were moved from their folders by typing * into the upper right-hand search bar, which simultaneously opens all files and shows contents individually listed. The individually listed files can then simply be copied into the working file.*
- mothur.exe
  - *The executable file begins outside the working file. It will move around.*
  - *uchime and vsearch files begin within the tegu_ed file.*

#I had to do these steps on my personal laptop as my drive would not work with 7zip on lab computer
In ISAAC cluster:

- go to email with file pathway
- open on demand
- files > home directory
- change directory, copy paste
- download necessary files to hard drive (.fastq.gz)

Extract .gz files into new folder using 7-zip into the fastq_extracted folder

- unzips the files, but still need to be opened (go into the file, put * in the search bar, and copy the opened files to the working folder)
- also copy and paste the uchime and vchime files into the working folder

Note: Commands with long processing times are noted with the time in GREEN.
#Note: Mothur is working when you see a blinking icon, when the command is fully executed and finished, it will have the output and mothur >

Mothur Pipeline

Now that the files are set up, we're ready to begin the Mothur pipeline.
Double-click on mothur.exe (located on My Passport) and run the following:
mothur > make.file(inputdir=tegu_wd, type=fastq, prefix=tegu_wd_fasta)

This only took a few minutes.
Here's the output from this command:

Setting input directories to:
tegu_wd\

3

Output File Names:
tegu_wd\tegu_wd_fasta.files

For ease of use, move the mothur.exe file into tegu_wd and work from within the file.
I closed and relaunched Mothur once the .exe file was moved into the working file.

Now we merge paired reads with the following command:
mothur > make.contigs(file= tegu_wd_fasta.files, processors=8)
##Double Check that the contig names are correct
This took about 3 hours and provided the following information:

Group count:
BLANK1 180
BLANK2 500
PCRBlank 140
T0C 88359
T0SB 31347
T0T 379241
T10C 575443
T10SB 33459
T10T 367446
T11C 121043
T11SB 8062
T11T 218686
T13C 232416
T13SB 10501
T13T 376386
T14C 305827
T14SB 339294
T14T 643946
T15C 39985

T15SB 7272
T15T 666698
T16C 111900
T16SB 305417
T16T 109394
T17C 332346
T17SB 129907
T17T 545883
T18C 591772
T18SB 87354
T18T 274864
T19C 325381
T19SB 145795
T19T 579027
T1C 657738
T1SB 969
T1T 354603
T20C 337081
T20SB 747016
T20T 121455
T21C 577301
T21SB 529946
T21T 322402
T22C 372227
T22SB 102131
T22T 499880
T23C 196624
T23SB 1048625
T23T 308240
T24C 41227
T24SB 681750
T24T 238048
T25C 529093
T25SB 249895
T25T 334866
T26C 535810
T26SB 325037
T26T 157219
T27C 409748
T27SB 455134
T27T 605444
T28C 64257
T28SB 9729
T28T 623565
T29C 312898

T29SB 3435
T29T 690695
T2C 317178
T2SB 232
T2T 277464
T30C 227710
T30SB 101732
T30T 363439
T31C 349612
T31SB 202225
T31T 201781
T3C 116182
T3SB 2555
T3T 516056
T4C 360931
T4SB 3545
T4T 266461
T5C 705632
T5SB 2298
T5T 305363
T6C 422425
T6SB 33065
T6T 296151
T7C 350242
T7SB 463238
T7T 448669
T8C 134481
T8SB 18763
T8T 178326
T9C 567209
T9SB 49368
T9T 108016

Total of all groups is 27819708

It took 3827 secs to process 27819708 sequences.

Output File Names:

tegu_wd_fasta.trim.contigs.fasta

tegu_wd_fasta.scrap.contigs.fasta

tegu_wd_fasta.contigs_report

tegu_wd_fasta.contigs.count_table

Running the following command shows what we have so far:

mothur> summary.seqs(fasta= tegu_wd_fasta.trim.contigs.fasta)

Using 20 processors.

	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	1	297	297	0	2	1
2.5%-tile:	1	307	307	0	4	695493
25%-tile:	1	308	308	0	4	6954928
Median:	1	308	308	0	5	13909855
75%-tile:	1	308	308	0	5	20864782
97.5%-tile:	1	552	552	2	21	27124216
Maximum:	1	600	600	138	300	27819708
Mean:	1	320	320	0	8	
# of Seqs:	27819708					

It took 335 secs to summarize 27819708 sequences.

Output File Names:
tegu_wd_fasta.trim.contigs.summary

If you know what you're doing, you can run the make.contigs command (above) and remove the primers at the same time [make.contigs (file=filename, oligos=oligofile.txt, processors=8)]. We only made the contigs (assembled the forward and reverse reads) and left the primers attached. There are several commands available to remove primers, one of which is pcr.seqs.

Mothur typically refers to primers and other adapters etc. as oligos and refers to them in that manner in their documentation. Oligos files are set up as text (.txt) files—it is easiest to do this is a text editor such as notepad. In this case the oligo file is named oligos16.txt and the contents are as follows:

forward GTGYCAGCMGCCGCGGTAA
reverse GGACTACNVGGGTWTCTAAT

Create a .txt file (in this case using notepad) named oligos16, copy/paste the primer sequences listed above, and save into the tegu_wd file.

The file may appear without the .txt extension. As long as the file type reads "Text Document" it should work fine.

#The following command was run to trim various primers off (ran overnight):

mothur > pcr.seqs(fasta= tegu_wd_fasta.trim.contigs.fasta, count=tegu_wd_fasta.contigs.count_table, oligos=oligos16.txt, pdiffs=2, rdiffs=2)

Running command: remove.seqs(accnos=tegu_wd_fasta.trim.contigs.bad.accnos, count=tegu_wd_fasta.contigs.count_table)
Removed 2518333 sequences from tegu_wd_fasta.contigs.count_table.

Output File Names:
tegu_wd_fasta.contigs.pick.count_table

/**/
It took 67905 secs to screen 27819708 sequences.

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.fasta
tegu_wd_fasta.trim.contigs.bad.accnos
tegu_wd_fasta.trim.contigs.scrap.pcr.fasta
tegu_wd_fasta.contigs.pcr.count_table

Running the following command shows what we have so far:
mothur > summary.seqs(fasta= tegu_wd_fasta.trim.contigs.pcr.fasta)

Using 20 processors.

	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	1	1	1	0	1	1
2.5%-tile:	1	252	252	0	4	632535
25%-tile:	1	253	253	0	4	6325344
Median:	1	253	253	0	5	12650688
75%-tile:	1	253	253	0	5	18976032
97.5%-tile:	1	254	254	1	8	24668841
Maximum:	1	559	559	102	175	25301375
Mean:	1	253	253	0	4	
# of Seqs:	25301375					

It took 383 secs to summarize 25301375 sequences.

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.summary

The next command will ordinarily be used to remove sequences with ANY ambiguous bases and sequences less than 50 and greater than 275 in the case of 16S processing. This is curated in the case of ITS, since sequences are of varying length, but not for 16S. General form of the command is as follows: screen.seqs(fasta=FILE.trim.contigs.pcr.fasta, count=FILE.contigs.pcr.groups, maxambig=0, maxlength=275). We instead run an edited version (immediately below).

The following command was run (this took about 10 min):

screen.seqs(fasta= tegu_wd_fasta.trim.contigs.pcr.fasta, count= tegu_wd_fasta.contigs.pcr.count_table, maxambig=0, minlength=50, maxlength=275)

It took 330 secs to screen 25301375 sequences, removed 1972505.

/**/
Running command: remove.seqs(accnos=tegu_wd_fasta.trim.contigs.pcr.bad.accnos.temp, count=tegu_wd_fasta.contigs.pcr.count_table)
Removed 1972505 sequences from tegu_wd_fasta.contigs.pcr.count_table.

Output File Names:
tegu_wd_fasta.contigs.pcr.pick.count_table

/**/

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.fasta
tegu_wd_fasta.trim.contigs.pcr.bad.accnos
tegu_wd_fasta.contigs.pcr.good.count_table

It took 924 secs to screen 25301375 sequences.

Running the following command takes a look at what we have so far:
summary.seqs(fasta= tegu_wd_fasta.trim.contigs.pcr.good.fasta)

Using 20 processors.

	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	1	50	50	0	2	1
2.5%-tile:	1	252	252	0	4	583222
25%-tile:	1	253	253	0	4	5832218
Median:	1	253	253	0	5	11664436
75%-tile:	1	253	253	0	5	17496653
97.5%-tile:	1	254	254	0	8	22745649
Maximum:	1	275	275	0	107	23328870
Mean:	1	252	252	0	4	
# of Seqs:	23328870					

It took 299 secs to summarize 23328870 sequences.

Output File Names:

tegu_wd_fasta.trim.contigs.pcr.good.summary

*# **Note:** we originally had 27,819,708 sequences. Now we have 23,328,870, which is 83.9% of what we started with.*

The following command was run. This will keep only the unique sequences:

unique.seqs(fasta=tegu_wd_fasta.trim.contigs.pcr.good.fasta, count=tegu_wd_fasta.contigs.pcr.good.count_table)

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.fasta
tegu_wd_fasta.trim.contigs.pcr.good.count_table
****This fasta file will be used in alignment.*

Running the following command shows what we have so far:

**Note: we used to rely on .names files in the older pipelines, but 1.48 and onward focus on count files, which are smaller and more efficient*.

summary.seqs(fasta= tegu_wd_fasta.trim.contigs.pcr.good.unique.fasta, count=tegu_wd_fasta.trim.contigs.pcr.good.count_table)

Using 20 processors.

	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	1	50	50	0	2	1
2.5%-tile:	1	252	252	0	4	583222
25%-tile:	1	253	253	0	4	5832218
Median:	1	253	253	0	5	11664436
75%-tile:	1	253	253	0	5	17496653
97.5%-tile:	1	254	254	0	8	22745649
Maximum:	1	275	275	0	107	23328870
Mean:	1	252	252	0	4	

of unique seqs: 1229856
total # of seqs: 23328870

It took 32 secs to summarize 23328870 sequences.

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.summary

#**********
Make reduced SILVA database:
#**********

Download SILVA database release 138.2 from the MOTHUR website, this is the newest version.
For me, 7 Zip works best with my drive on my personal laptop, so I did these steps there

Steps for downloading the SILVA database:

- Go to the MOTHUR website and find the release you want (v138.2, in this case), and take the first download option ("Full-length Sequences and Taxonomy References").
    - I moved this download onto my hard drive (but NOT the working folder) for ease of locating it.
- Once the download is complete, open 7-zip (file extractor) and from within 7-zip select the download file. Extract the file. A new folder will be created with the same name. Open this folder and it should contain a file with the same name.
- Go back into 7-zip and extract this file (essentially this is a double extraction).
- NOW you should have a folder that contains five files: a README markdown file, silva.nr_v138_2.align, silva.nr_v138_2.tax, ._ silva.nr_v138_2.align, and ._silva.nr_v138_2.tax. I am not really sure what the "._" files are, as they were not in prior versions, so I am avoiding them.
- Copy silva.nr_v138_2.align and silva.nr_v138_2.tax into the working folder.

Steps for creating the E. coli reference file:

- Go to the NCBI and under the DNA tabs hunt around until you find an E. coli 16S reference sequence.
    - This one has been used in previous runs and is what I used: (accession number) NR_024570.1 Escherichia coli strain U 5/41 16S ribosomal RNA, partial sequence
- Download the sequence and save it as a .fasta file (named ecoli_ref here) to the working file.

#1: Find where our primers fall in the started e. coli 16S rRNA sequence from NCBI and remove the flanking regions.

mothur > pcr.seqs(fasta=ecoli_ref.fasta, oligos=oligos16.txt)

Using 20 processors.
1
[NOTE]: no sequences were bad, removing ecoli_ref.bad.accnos

It took 0 secs to screen 1 sequences.

Output File Names:
ecoli_ref.pcr.fasta

#2: Align the trimmed E. coli sequence to the SILVA database to find what regions in the database are covered.

mothur > align.seqs(fasta= ecoli_ref.pcr.fasta, reference=silva.nr_v138_2.align)

Output File Names:

ecoli_ref.pcr.align
ecoli_ref.pcr.align_report
Using 20 processors.

Reading in the silva.nr_v138_2.align template sequences... DONE.
It took 480 to read 164296 sequences.

Aligning sequences from ecoli_ref.pcr.fasta ...
Reducing processors to 1.
1
It took 1 secs to align 1 sequences.

It took 23 seconds to align 1 sequences.

Output File Names:
ecoli_ref.pcr.align
ecoli_ref.pcr.align_report

#3: Show where our primers start and stop in the SILVA database.
mothur > summary.seqs(fasta= ecoli_ref.pcr.align)
Using 20 processors.

	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	13862	23444	253	0	6	1
2.5%-tile:	13862	23444	253	0	6	1
25%-tile:	13862	23444	253	0	6	1
Median:	13862	23444	253	0	6	1
75%-tile:	13862	23444	253	0	6	1
97.5%-tile:	13862	23444	253	0	6	1
Maximum:	13862	23444	253	0	6	1
Mean:	13862	23444	253	0	6	
# of Seqs:	1					

It took 3 secs to summarize 1 sequences.

Output File Names:
ecoli_ref.pcr.summary

#4: Trim the Silva database to only the region our primers cover to reduce the database (this makes it faster and easier to work with.

mothur > pcr.seqs(fasta= silva.nr_v138_2.align, start=13862, end=23444, keepdots=F, processors=8)
[NOTE]: no sequences were bad, removing silva.nr_v138_2.bad.accnos

It took 116 secs to screen 164296 sequences.

Output File Names:
silva.nr_v138_2.pcr.align

#5: Rename the file.
mothur > rename.file(input= silva.nr_v138_2.align, new=silva.nr_v138_2.v4.fasta)
Current files saved by mothur:
accnos=tegu_wd_fasta.trim.contigs.pcr.bad.accnos
fasta=silva.nr_v138_2.pcr.align
oligos=oligos16.txt
count=tegu_wd_fasta.trim.contigs.pcr.good.count_table
processors=8
summary=ecoli_ref.pcr.summary

#6: Perform summary.
mothur > summary.seqs(fasta=silva.nr_v138_2.v4.fasta)

Using 8 processors.

	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	1044	31161	920	0	4	1
2.5%-tile:	1044	43116	1393	0	5	4108
25%-tile:	1044	43116	1438	0	5	41075
Median:	1044	43116	1455	0	6	82149
75%-tile:	1044	43116	1474	0	6	123223
97.5%-tile:	1044	43116	1745	2	7	160189
Maximum:	1120	43116	3661	5	24	164296
Mean:	1044	43115	1477	0	5	
# of Seqs:	164296					

It took 96 secs to summarize 164296 sequences.

Output File Names:
silva.nr_v138_2.v4.summary

#**********
Done with manipulating the SILVA database. Now we align our sequences.

#**********

Align unique sequences to reduced reference (use the fasta file made during the unique.seqs command, just prior to making the count table -- this is indicated with a RED note earlier in the text) THIS RAN OVER NIGHT.

align.seqs(fasta= tegu_wd_fasta.trim.contigs.pcr.good.unique.fasta, reference=silva.nr_v138_2.v4.fasta)

It took 17669 secs to align 1229856 sequences.

[WARNING]: 27718 of your sequences generated alignments that eliminated too many bases, a list is provided in tegu_wd_fasta.trim.contigs.pcr.good.unique.flip.accnos.
[NOTE]: 13514 of your sequences were reversed to produce a better alignment.

It took 17728 seconds to align 1229856 sequences.

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.align
tegu_wd_fasta.trim.contigs.pcr.good.unique.align_report
tegu_wd_fasta.trim.contigs.pcr.good.unique.flip.accnos

Run the summary using the count table generated just before manipulating the SILVA database (this is indicated with a RED note earlier in the text). THIS TOOK ALMOST AN HOUR

mothur > summary.seqs(fasta= tegu_wd_fasta.trim.contigs.pcr.good.unique.align, count= tegu_wd_fasta.trim.contigs.pcr.good.count_table)

Using 20 processors.

	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	0	0	0	0	1	1
2.5%-tile:	13862	23443	252	0	4	583222
25%-tile:	13862	23444	253	0	4	5832218
Median:	13862	23444	253	0	5	11664436
75%-tile:	13862	23444	253	0	5	17496653
97.5%-tile:	13862	23444	254	0	8	22745649
Maximum:	43116	43116	275	0	83	23328870
Mean:	13894	23449	252	0	4	

of unique seqs: 1229856
total # of seqs: 23328870

It took 2657 secs to summarize 23328870 sequences.

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.summary

#Set start and end based on summary seqs to see were majority of sequences start and end. This took a little under an hour

mothur> screen.seqs(fasta=tegu_wd_fasta.trim.contigs.pcr.good.unique.align, count=tegu_wd_fasta.trim.contigs.pcr.good.count_table, summary=tegu_wd_fasta.trim.contigs.pcr.good.unique.summary, start=13862**, end=**23444**, maxhomop=8)**

It took 3066 secs to screen 1229856 sequences, removed 330281.

/**/
Running command: remove.seqs(accnos=tegu_wd_fasta.trim.contigs.pcr.good.unique.bad.accnos.temp, count=tegu_wd_fasta.trim.contigs.pcr.good.count_table)
Removed 4181819 sequences from tegu_wd_fasta.trim.contigs.pcr.good.count_table.

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.pick.count_table

/**/

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.summary
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.align
tegu_wd_fasta.trim.contigs.pcr.good.unique.bad.accnos
tegu_wd_fasta.trim.contigs.pcr.good.good.count_table

It took 4027 secs to screen 1229856 sequences.

mothur > summary.seqs(fasta=current, count=current)
This took about half an hour
Using tegu_wd_fasta.trim.contigs.pcr.good.good.count_table as input file for the count parameter.
Using tegu_wd_fasta.trim.contigs.pcr.good.unique.good.align as input file for the fasta parameter.

Using 20 processors.

	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	10273	23444	209	0	3	1
2.5%-tile:	13862	23444	252	0	4	478677

25%-tile: 13862 23444 253 0 4 4786763
Median: 13862 23444 253 0 5 9573526
75%-tile: 13862 23444 253 0 5 14360289
97.5%-tile: 13862 23444 254 0 8 18668375
Maximum: 13862 26158 275 0 8 19147051
Mean: 13861 23444 253 0 4
of unique seqs: 899575
total # of seqs: 19147051

It took 1681 secs to summarize 19147051 sequences.

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.summary

Remove overhangs This took about 40 min
mothur > filter.seqs(fasta= tegu_wd_fasta.trim.contigs.pcr.good.unique.good.align, vertical=T, trump=.)
It took 2415 secs to filter 899575 sequences.

Length of filtered alignment: 770
Number of columns removed: 49230
Length of the original alignment: 50000
Number of sequences used to construct filter: 899575

Output File Names:
tegu_wd_fasta.filter
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.fasta

Identify unique sequences
mothur > unique.seqs(fasta=current, count=current)

Using tegu_wd_fasta.trim.contigs.pcr.good.good.count_table as input file for the count parameter.
Using tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.fasta as input file for the fasta parameter.

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.fasta
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.count_table

#pre-cluster sequences (split by group and sort by abundance; seq within 2 nt difference will get merged).
mothur > pre.cluster(fasta=current, count=current, diffs=2)

Using tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.count_table as input file for the count parameter.
Using tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.fasta as input file for the fasta parameter.

Using 20 processors.

Running get.seqs:
Selected 398095 sequences from tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.fasta.
/**/
It took 819 secs to run pre.cluster.

Using 20 processors.

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.fasta
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.count_table
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.BLANK2.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T0C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T0SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T0T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T10C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T10SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T10T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T11C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T11SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T11T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T13C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T13SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T13T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T14C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T14SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T14T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T15C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T15SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T15T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T16C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T16SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T16T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T17C.map

tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T17SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T17T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T18C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T18SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T18T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T19C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T19SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T19T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T1C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T1SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T1T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T20C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T20SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T20T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T21C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T21SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T21T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T22C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T22SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T22T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T23C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T23SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T23T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T24C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T24SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T24T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T25C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T25SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T25T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T26C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T26SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T26T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T27C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T27SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T27T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T28C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T28SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T28T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T29C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T29SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T29T.map

tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T2C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T2SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T2T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T30C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T30SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T30T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T31C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T31SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T31T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T3C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T3SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T3T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T4C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T4SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T4T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T5C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T5SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T5T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T6C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T6SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T6T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T7C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T7SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T7T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T8C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T8SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T8T.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T9C.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T9SB.map
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.T9T.map

Remove chimeras with vsearch (dereplicate=t, defaults are dereplicate=t, dups=t)
mothur > chimera.vsearch(fasta=
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.fasta, count=
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.count_table, dereplicate=T)

Removing chimeras from your input files:
/**/
Running command:
remove.seqs(fasta=tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.fasta,

accnos=tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.accnos)
Removed 192585 sequences from tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.fasta.

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.fasta

/**/

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.count_table
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.chimeras
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.accnos
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.fasta

#perform summary
summary.seqs(fasta=current, count=current)

Using tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.count_table as input file for the count parameter.
Using tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.fasta as input file for the fasta parameter.

Using 20 processors.

	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	1	709	209	0	3	1
2.5%-tile:	1	770	252	0	4	468326
25%-tile:	1	770	253	0	4	4683260
Median:	1	770	253	0	5	9366519
75%-tile:	1	770	253	0	5	14049778
97.5%-tile:	1	770	254	0	8	18264711
Maximum:	5	770	275	0	8	18733036
Mean:	1	769	253	0	4	

of unique seqs: 205510
total # of seqs: 18733036

It took 5 secs to summarize 18733036 sequences.

Output File Names:

tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.summary

Remove chimeras from fasta file
mothur > remove.seqs(fasta=
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.fasta, accnos=
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.accnos)

Removed 192585 sequences from
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.fasta.

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.fasta

#running command
summary.seqs(fasta=current, count=current)

Using
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.count_table as input file for the count parameter.
Using tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.fasta as input file for the fasta parameter.

Using 20 processors.

	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	1	709	209	0	3	1
2.5%-tile:	1	770	252	0	4	468326
25%-tile:	1	770	253	0	4	4683260
Median:	1	770	253	0	5	9366519
75%-tile:	1	770	253	0	5	14049778
97.5%-tile:	1	770	254	0	8	18264711
Maximum:	5	770	275	0	8	18733036
Mean:	1	769	253	0	4	
# of unique seqs:	205510					
total # of seqs:	18733036					

It took 7 secs to summarize 18733036 sequences.

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.summary

The original number of sequences was 27,819,708 sequences. After trimming and removing ambiguous bases we had 23,328,870. Between clustering a removing chimera we have 18,733,036. This is about 71% of what we had originally. Unique sequences have been reduced from 1,229,856 to 205,510 which is about 17% of what we had originally.

#classify sequences (takes about 1 hour)

mothur > classify.seqs(fasta=current, count=current, reference=silva.nr_v138_2.align, taxonomy=silva.nr_v138_2.tax, cutoff=80)

Using tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.count_table as input file for the count parameter.
Using tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.fasta as input file for the fasta parameter.

It took 1540 secs to classify 205510 sequences.

It took 15 secs to create the summary file for 205510 sequences.

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.nr_v138_2.wang.taxonomy
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.nr_v138_2.wang.tax.summary

#remove non-bacterial sequences

mothur > remove.lineage(fasta=current, count=current, taxonomy= tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.nr_v138_2.wang.taxonomy, taxon=Chloroplast=Mitochondria-unknown-Archaea-Eukaryota)

Using tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.count_table as input file for the count parameter.
Using tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.fasta as input file for the fasta parameter.

/**/
Running command:
remove.seqs(accnos=tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.nr_v138_2.wang.accnos, count=tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.count_table, fasta=tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.fasta)
Removed 2408 sequences from tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.fasta.

Removed 95061 sequences from
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.count_table.

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.pick.fasta
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.pick.count_table

/**/

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.nr_v138_2.wang.pick.taxonomy
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.nr_v138_2.wang.accnos
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.pick.count_table
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.pick.fasta

mothur > summary.tax(taxonomy=current, count=current)

Using
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.pick.count_table as input file for the count parameter.
Using
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.nr_v138_2.wang.pick.taxonomy as input file for the taxonomy parameter.

It took 8 secs to create the summary file for 18637975 sequences.

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.nr_v138_2.wang.pick.tax.summary

#cluster sequences into OTUs based on distance (97% sequence similarity)-took about 4.5 hours
mothur > dist.seqs(fasta=current, cutoff=0.03)

It took 15466 secs to find distances for 203102 sequences. 10349896 distances below cutoff 0.03.

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.pick.dist

We move on to clustering sequences based on our calculated distance matrix, and with a cutoff of 0.03. (For fungal sequences we had to use greedy clustering, using a cutoff of 0.05 and method=agc (this is a VSEARCH clustering option because contigs are of differing lengths, therefore no dist matrix can be calculated. For bacterial sequences we use a different clustering option.)

#this is using the default opticlust method; default mcc metric; default 0.03 cutoff
mothur > cluster(column=
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.pick.dist,
count=current)

Using
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.pick.count_table
as input file for the count parameter.

Using 20 processors.

You did not set a cutoff, using 0.03.

Clustering tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.pick.dist

It took 233 seconds to cluster

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.pick.opti_mcc.list
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.pick.opti_mcc.steps
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.pick.opti_mcc.sensspec

Make the OTU table.
mothur >
make.shared(list=tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.pick.opti_mcc.list, count=current)

Using
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.pick.count_table
as input file for the count parameter.
0.03

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.pick.opti_mcc.shared

And now classify the OTUs based SILVA database taxonomy
mothur >
classify.otu(list=tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.pick.opti_mcc.list, count=current, taxonomy=current, label=0.03)

Using
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.denovo.vsearch.pick.count_table
as input file for the count parameter.

Using tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.nr_v138_2.wang.pick.taxonomy as input file for the taxonomy parameter.
0.03

Output File Names:
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.pick.opti_mcc.0.03.cons.taxonomy
tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.pick.opti_mcc.0.03.cons.tax.summary

Shared (from make.shared) and Taxonomy (from classify.otu) are now used for downstream processing in Phyloseq.

tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.pick.opti_mcc.shared

tegu_wd_fasta.trim.contigs.pcr.good.unique.good.filter.unique.precluster.pick.pick.opti_mcc.0.03.cons.taxonomy

For ease of use, these files were copied to (and renamed in) file “Mothur Pipeline Output”. File names are as follows:

tegu16s2.shared
tegu16s2.taxonomy